

On Self-Similar Viscosity Models for Carreau Boundary Layer Flows

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- ① Motivation: boundary layers in coating-flow applications
- ② Rheological models: Newtonian, power-law, Carreau
- ③ Formulation and (pseudo-)similarity
- ④ **New** power-law Sakiadis solutions for $\alpha < \frac{1}{2}$ and $\alpha > 1$
- ⑤ Wall shear: comparing self-similar predictions to Carreau
- ⑥ A simple criterion for combining similarity solutions
- ⑦ Conclusions

Motivation

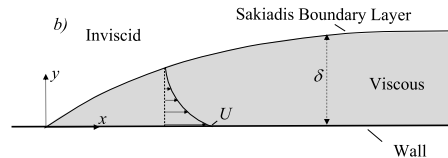
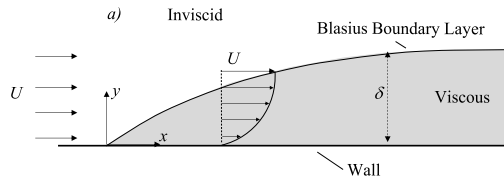
Boundary layers are thin, viscosity-dominated regions adjacent to a wall.

Two canonical flat-wall problems:

- **Blasius (1908)** — fluid moves over a stationary wall.
- **Sakiadis (1961)** — wall moves through stationary fluid.

Sakiadis flows are directly relevant to *liquid film coating*, fiber drawing, and other industrial processes.

Real coating fluids are *non-Newtonian* — their viscosity depends on shear rate.

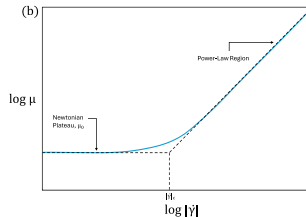
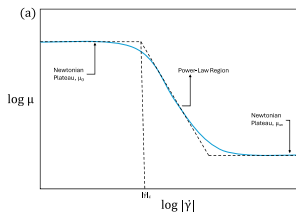


Why Care About the Full Rheological Profile?

A single boundary layer flow accesses a *wide* range of strain-rate magnitudes:

- Near the wall: $|\dot{\gamma}|$ is large.
- Far from the wall: $|\dot{\gamma}| \rightarrow 0$.
- Far downstream: $|\dot{\gamma}|$ decreases as the boundary layer grows.

So the *whole* rheological curve can matter — not just the power-law region.



Left: shear-thinning. **Right:** shear-thickening.

Solid: Carreau. Dashed: power-law and Newtonian asymptotes.

Three Viscosity Models

Power-law (Ostwald–de Waele).

$$\mu = K |\dot{\gamma}|^{\alpha-1}.$$

Captures shear-thinning ($\alpha < 1$) or shear-thickening ($\alpha > 1$); admits similarity solutions.

But: $\mu \rightarrow \infty$ or $\mu \rightarrow 0$ in unphysical limits.

Carreau (with $\mu_\infty = 0$).

$$\mu = \mu_0 (1 + (\lambda |\dot{\gamma}|)^2)^{(\alpha-1)/2}.$$

Smoothly connects a Newtonian plateau at low $|\dot{\gamma}|$ to power-law behavior at high $|\dot{\gamma}|$.

Cost: no exact similarity solution.

Connecting them: $K = \mu_0 \lambda^{\alpha-1}$, $|\dot{\gamma}|_c = 1/\lambda$.

Central question

How well do the simpler, *self-similar* Newtonian and power-law solutions describe the boundary layer of a Carreau fluid, and *when* can they be used in place of it?

Metric: wall shear rate $\bar{\tau} = \tau/(\rho U^2)$.

Two contributions:

- 1 New self-similar Sakiadis power-law solutions for $\alpha < 1/2$ and $\alpha > 1$ — previously believed not to exist.
- 2 A quantitative comparison of wall-shear predictions across models, leading to a dimensionless criterion for combining similarity solutions to approximate the Carreau result.

Boundary-Layer Formulation

Streamwise velocity u_x , transverse u_y , with $\tau_{yx} = \mu(|\dot{\gamma}|) \partial u_x / \partial y$:

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0, \quad \rho \left(u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = \frac{\partial \tau_{yx}}{\partial y}.$$

Location	Blasius	Sakiadis
$y = 0$	$u_x = 0, u_y = 0$	$u_x = U, u_y = 0$
$y \rightarrow \infty$	$u_x \rightarrow U$	$u_x \rightarrow 0$

Pressure plays no role; boundary layer is driven entirely by relative wall–fluid motion.

Power-Law Similarity Reduction

Substituting the standard similarity variables for a power-law fluid,

$$\eta = y \left(\frac{\rho U^{2-\alpha}}{K_X} \right)^{1/(\alpha+1)}, \quad f_p(\eta; \alpha) = \frac{\eta \psi}{U y},$$

reduces the PDE to a single ODE:

$$\alpha(\alpha + 1) f_p''' + f_p f_p'' |f_p''|^{1-\alpha} = 0,$$

with the BCs $f_p(0) = 0$, $f_p'(0) = 0$, $f_p'(\infty) \rightarrow 1$ (Blasius) or $f_p'(0) = 1$, $f_p'(\infty) \rightarrow 0$ (Sakiadis).

Wall shear rate (dimensional):

$$\tau_p = K \left(U |f_p''(0; \alpha)| \left(\frac{\rho U^{2-\alpha}}{K_X} \right)^{1/(\alpha+1)} \right)^\alpha.$$

No exact similarity exists, but following *Griffiths et al. (2016)* we use the *Newtonian* similarity variable; the x -dependence survives only as a parameter:

$$\delta = \frac{\lambda^2 U^3 \rho}{\mu_0 x} = \frac{\text{De}^2}{\text{Re}_{n,x}},$$

giving a *one-parameter family* of ODEs:

$$(1 + \alpha \delta f_c''^2) (1 + \delta f_c''^2)^{(\alpha-3)/2} f_c''' + \frac{1}{2} f_c f_c'' = 0.$$

- One ODE solve per streamwise location x .
- δ acts as an effective relaxation time — shrinks downstream as the boundary layer grows.
- Wall shear: $\tau_c = \sqrt{\rho \mu_0 U^3 / x} f_c''(0; \delta) (1 + \delta f_c''(0; \delta)^2)^{(\alpha-1)/2}$.

For the Sakiadis power-law problem, prior work (Fox et al. 1969; Naghshineh et al. 2023) reports *no solution* for

$$\alpha < \frac{1}{2} \quad \text{or} \quad \alpha > 1.$$

The argument assumed an asymptotic form $f_p \sim C + h(\eta)$ with $h \ll C$ as $\eta \rightarrow \infty$. The dominant balance gives

$$f_p \sim C + D(1 + E\eta)^{(2\alpha-1)/(\alpha-1)}, \quad \frac{1}{2} < \alpha < 1,$$

which fails self-consistency outside this window.

Resolution

The boundary condition $f_p'(\infty) \rightarrow 0$ does *not* require $f_p \rightarrow C$. Allowing f_p to grow algebraically opens up new admissible solutions.

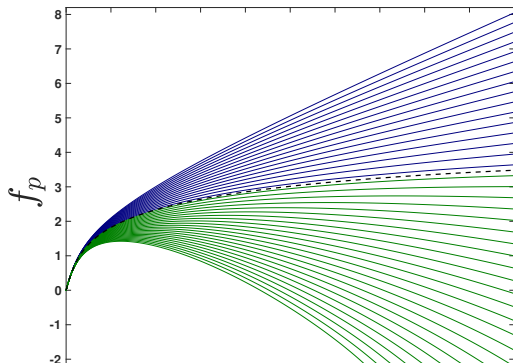
New Asymptotic Form for $\alpha < 1/2$

We propose

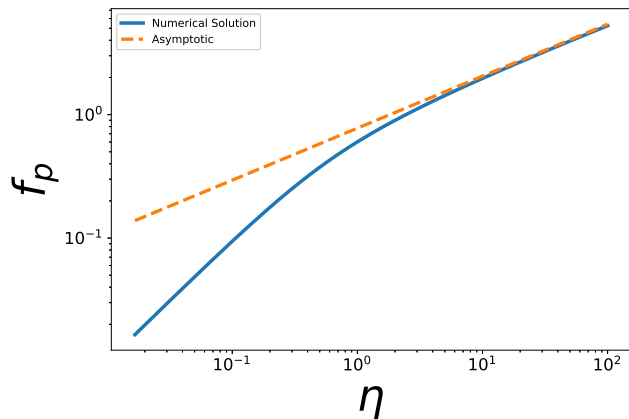
$$f_p(\eta; \alpha) \sim H(\eta + J)^p, \quad p = \frac{1 - 2\alpha}{2 - \alpha}, \quad H = \left(\frac{\alpha(\alpha + 1)(2 - p)}{p^{1-\alpha}(1 - p)^{1-\alpha}} \right)^{1/(2-\alpha)}.$$

Treating the BVP as an IVP in $\kappa = f_p''(0; \alpha)$:

- $\kappa > \kappa_p$: monotone, unbounded.
- $\kappa < \kappa_p$: nonmonotone, eventually decreases.
- $\kappa = \kappa_p$: a separatrix — the unique trajectory smoothly approaching $f_p' \rightarrow 0$.



Verifying the Asymptotic



Log-log plot of the separatrix solution ($\alpha = 0.1$).

Solid: numerical solution.

Dashed: predicted $f_p \sim H(\eta + J)^p$ (with $J = 0$).

Numerics matches the asymptotic form, supporting the new branch of solutions for $\alpha < 1/2$.

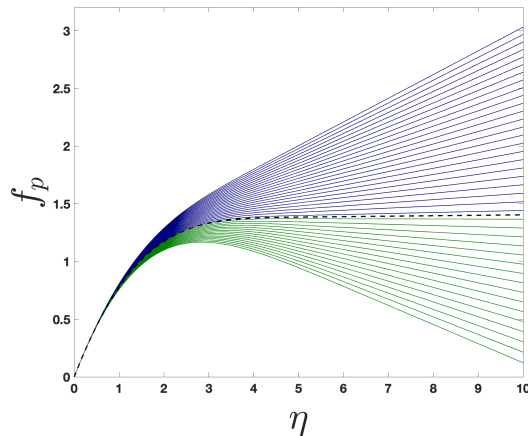
Shear-Thickening Sakiadis ($\alpha > 1$)

A separatrix structure exists *again* — but now with a qualitatively different form:

$$f_p(\eta; \alpha) = C_2 \quad \text{for all } \eta \geq C_1.$$

So u_x is identically zero outside a finite-width region: a *finite-thickness boundary layer*.

The ODE term $|f_p''|^{2-\alpha}$ is **not** Lipschitz for $\alpha > 1$, so Picard–Lindelöf fails — the IVP can have non-unique continuations, allowing this singular structure.



Shooting solutions for $\alpha = 1.4$.

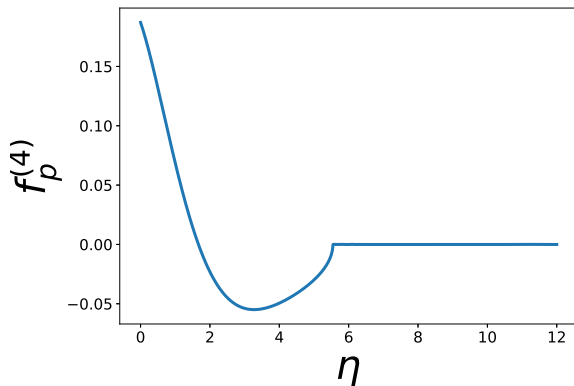
C^4 but not C^5 at the Boundary-Layer Edge

At the edge $\eta = C_1$, the new Sakiadis solution is

- continuous in f_p, f'_p, f''_p ;
- **discontinuous in $f_p^{(5)}$.**

Exactly analogous to the Blasius shear-thickening result of Zhizhin & Ufimtsev (1978) and Pavlov et al. (1982).

To our knowledge, this is the *first* identification of this structure for the Sakiadis problem.



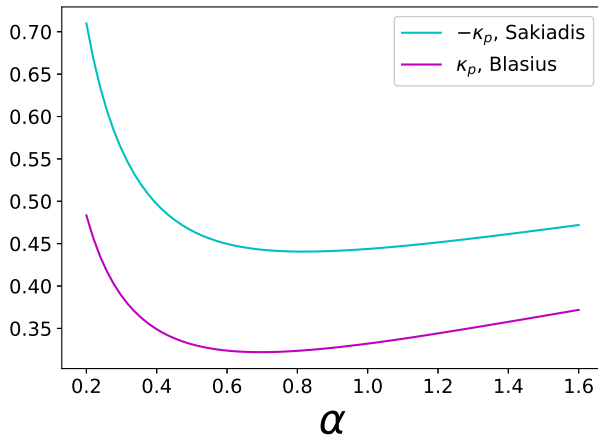
Fourth derivative of the Sakiadis separatrix solution, $\alpha = 1.4$.

Wall-Shear Coefficient $\kappa_p(\alpha)$

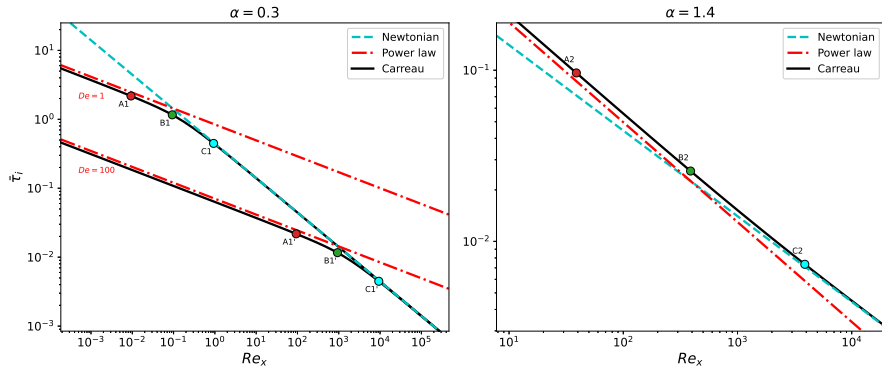
The new Sakiadis solutions glue smoothly onto the previously known range $1/2 < \alpha \leq 1$:

- $\kappa_p(\alpha)$ is *continuous* across $\alpha = 1/2$ and $\alpha = 1$.
- Sakiadis and Blasius curves are qualitatively similar.

We will see in a moment that these κ_p values track the Carreau wall shear in the high-shear (low- $\text{Re}_{n,x}$) limit.



Wall-Shear Comparison: All Three Models



Sakiadis wall shear vs. $Re_{n,x}$ at fixed De , for shear-thinning ($\alpha = 0.3$, multiple De) and shear-thickening ($\alpha = 1.4$). **Power-law line** dominates near the leading edge; **Newtonian line** dominates far downstream; Carreau bridges the two.

Across all α , De examined (Sakiadis and Blasius):

- As $Re_{n,x} \rightarrow \infty$ (far downstream): $\delta \rightarrow 0$, the Carreau ODE reduces to the Newtonian one, and

$$\bar{\tau}_c \sim \bar{\tau}_n.$$

- As $Re_{n,x} \rightarrow 0$ (near leading edge):

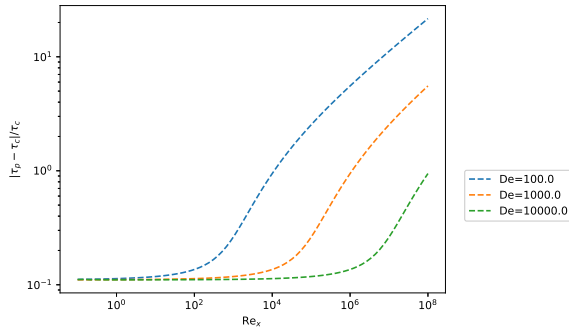
$$\bar{\tau}_c \sim (1 - c(\alpha))\bar{\tau}_p.$$

The power-law line offers an excellent *slope* match, but with a finite vertical offset.

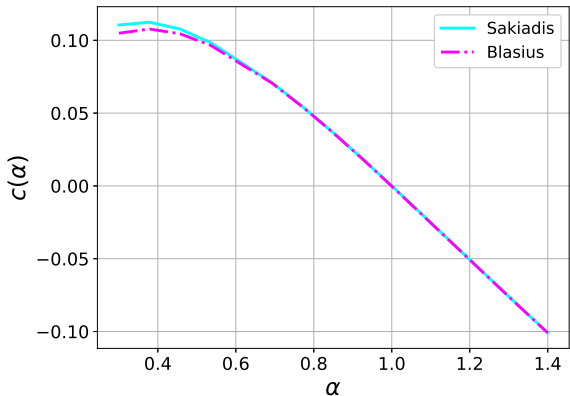
Two key questions follow:

- ① How big is the correction $c(\alpha)$, and where does it come from?
- ② Can we predict where Carreau transitions between regimes?

The Correction $c(\alpha)$



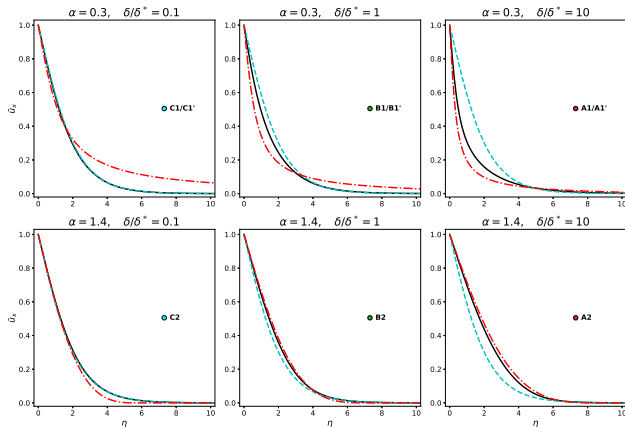
Relative error $|\tau_p - \tau_c|/\tau_c$ for $\alpha = 0.3$, Sakiadis flow.



$c(\alpha)$ for Sakiadis and Blasius.

- Error becomes *independent of De* as $Re_{n,x} \rightarrow 0$ — justifying $c = c(\alpha)$.
- For $\alpha \in [0.3, 1.4]$: $|c(\alpha)| \lesssim 11\%$.

Where Does $c(\alpha)$ Come From? — Velocity Profiles



Integrated form of the BL momentum equation:

$$\bar{\tau}_i = \frac{d}{dx} \int_0^\infty u_x (U - u_x) dy.$$

The *whole* velocity field affects $\bar{\tau}$, not just the slope at the wall.

In the power-law regime: slopes at the wall agree, but profiles *differ away from the wall* — this is exactly the source of the residual $c(\alpha)$.

Setting $\bar{\tau}_p = \bar{\tau}_n$ gives the Newtonian–power-law crossover:

$$\delta^* = \left(\frac{\kappa_p^\alpha}{\kappa_n} \right)^{2(1+\alpha)/(1-\alpha)}$$

- Both $\kappa_p(\alpha)$ and κ_n come from *self-similar* solutions only.
- Over $\alpha \in (0, 2)$, δ^* ranges only from ~ 14 to ~ 20 — remarkably insensitive to α .
- Recipe:
 - $\delta/\delta^* \gg 1$ (near leading edge): use power-law $\bar{\tau}_p$.
 - $\delta/\delta^* \ll 1$ (far downstream): use Newtonian $\bar{\tau}_n$.
- Total drag on the plate $\Rightarrow$ split the integral at $\delta = \delta^*$ — no Carreau ODE solve needed.

Engineering recipe (no Carreau solver required)

- 1 Look up $\kappa_p(\alpha)$ and $\kappa_n = f_p''(0; 1)$ from self-similar solutions.
- 2 Compute δ^* from the formula on the previous slide.
- 3 Convert to a critical x^* via $\delta = \lambda^2 U^3 \rho / (\mu_0 x)$.
- 4 Use $\bar{\tau}_p$ upstream of x^* and $\bar{\tau}_n$ downstream of x^* .

Errors with respect to the full Carreau solution:

- $\sim 11\%$ at most in the power-law regime, attributable to $c(\alpha)$.
- Asymptotically zero in the Newtonian regime.
- Intermediate (transition) region: largest error — the place to use the Carreau ODE if accuracy demands it.

Denier & Dabrowski (2004): Blasius shear-thickening solutions lose smoothness because the BL approximation itself breaks down where viscosity is small. They introduce a “viscous diffusion layer” to recover continuity.

Our finding: restoring a Newtonian plateau at low shear (i.e., the Carreau model) yields a completely smooth solution *without* modifying the boundary-layer equation.

And: the new Sakiadis power-law solutions presented here mirror the well-known Blasius singular structure for $\alpha > 1$, completing the parallel between the two flows.

- ① **New Sakiadis power-law solutions** for $\alpha < 1/2$ (algebraic decay, $f_p \sim H(\eta + J)^p$) and $\alpha > 1$ (finite-width boundary layer, C^4 but not C^5). Wall-shear values $\kappa_p(\alpha)$ vary continuously with α .
- ② **Wall-shear of a Carreau fluid** is well-approximated by self-similar models in their regimes:
 - Newtonian asymptote — exact match as $\text{Re}_{n,x} \rightarrow \infty$.
 - Power-law asymptote — offset by a quantifiable $c(\alpha)$ ($\lesssim 11\%$ for $\alpha \in [0.3, 1.4]$).
- ③ **The residual** $c(\alpha)$ originates from differences in velocity profiles *away from the wall*, consistent with the integrated BL momentum balance.
- ④ **A simple criterion** δ^* tells you where to switch between asymptotes — giving accurate Carreau wall-shear estimates without solving the Carreau ODE family.

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Thank you — questions?